



ST BENEDICT'S

SUBJECT	CHEMISTRY EXAMINATION
GRADE	11
EXAMINER	Mr McCoy
MODERATOR	Mr Ntombela
DATE	11 November 2024
MARKS	135
DURATION	2 hours

%

LEARNER NAME:

TEACHER:	GOVENDER	HILDER	MCCOY	NTOMBELA
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PLEASE READ THE FOLLOWING INSTRUCTIONS CAREFULLY

1. This question paper consists of 16 pages and a separate Data Sheet and Periodic Table. Please check that your question paper is complete.
2. Read the questions carefully.
3. Answer **ALL** the questions on the question paper and hand it in at the end of the assessment.
4. Remember to write your name and surname in the space provided above AND circle your teacher's name
5. Use the data and formulae sheet whenever necessary.
6. Show your working in all calculations.
7. Units need not be included in the working of calculations, but appropriate units should be shown in the answer.
8. Answers must be expressed in decimal format, not left as proper fractions.
9. Where appropriate, express answers to TWO decimal places.
10. It is in your own interest to write legibly and to present your work neatly.
11. A Blank page is included at the end of the paper. If you run out of space for a question, use this space. Clearly indicate the question number of your answer should you use this extra space.

Question	1	2	3	4	5	6	Total
Mark							
Moderated Mark							
Question Total	22	26	25	15	13	34	135

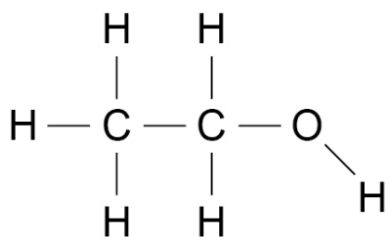
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Learner's Signature:

Teacher's Signature:

QUESTION 1**22 MARKS**

- 1.1 Complete the diagram to show how one molecule of **ammonia** can form a hydrogen bond with one molecule of ethanol.
Include all **lone pairs** of electrons;
and **partial charges** on atoms involved in the hydrogen bond. (4)



- 1.2 **Table 1** shows the electronegativity values of atoms of some elements.

Table 1

Atom	H	C	N	O	Br
Electronegativity	2,1	2,5	3,0	3,5	2,8

- 1.2.1 Define the term *electronegativity*. (2)

- 1.2.2 Deduce the two atoms from **Table 1** that will form the most polar bond. (1)

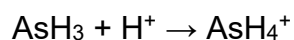
- 1.2.3 The C–Br bond is polar
Explain why CBr₄ is not a polar molecule. (2)

- 1.2.4 Suggest, in terms of the intermolecular forces for each compound, why CBr_4 has a **higher** boiling point than CHBr_3 . (5)

- 1.2.5 Name the physical form of carbon that can conduct electricity. (1)

- 1.2.6 Explain how it is possible for carbon to conduct electricity in this form. (2)

- 1.3 The equation for a reaction is

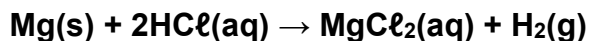


- 1.3.1 Classify the specific bonds between the atoms in AsH_3 . (2)

- 1.3.2 A dative covalent bond forms in this reaction. Use a Lewis diagram to show the bonding in AsH_4^+ . Indicate the electrons of each element differently. (3)

QUESTION 2**26 MARKS**

A student conducts an experiment to investigate the effect of different concentrations of HCl on the reaction rate with excess magnesium. The reaction is as follows:



- They use 0,5 mol.dm⁻³, 1 mol.dm⁻³, and 2 mol.dm⁻³ solutions of HCl, keeping the volume of acid and mass of magnesium constant.
- The volume of hydrogen gas produced is measured every 10 seconds.

2.1 Define the term *reaction rate* for a chemical reaction. (2)

2.2 Use collision theory to explain how the concentration of hydrochloric acid affects the rate of the reaction. (4)

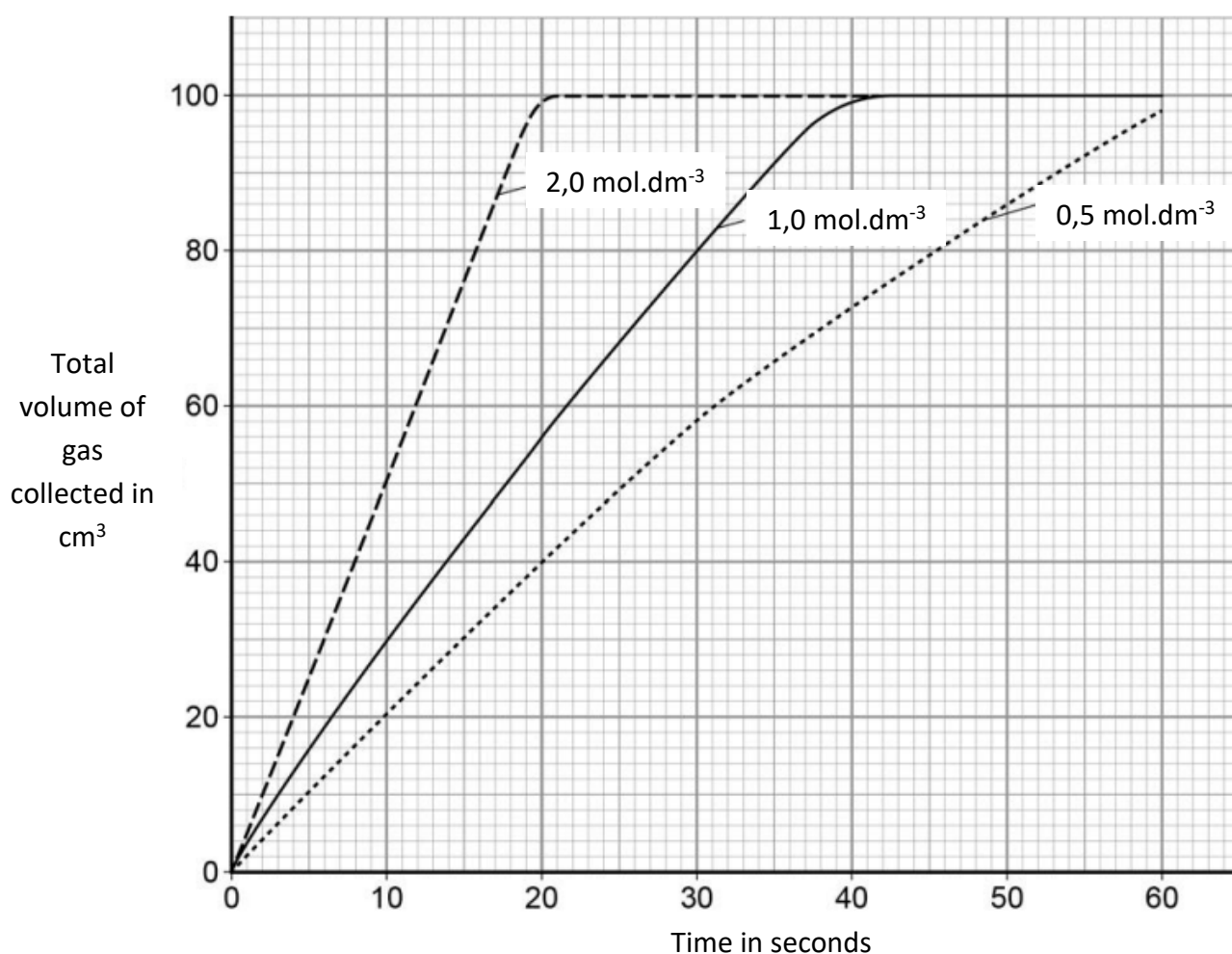
2.3 Briefly explain why an increase in the temperature of the reaction mixture would increase the reaction rate. (2)

2.4 Sketch a Boltzmann Distribution Curve for a mixture at high and low temperature, to compare the effect of temperature on reaction rate. Label your graphs. (3)



- 2.5 The student wishes to verify that the experiment is accurate. List two control variables they should use to ensure a fair comparison. (2)

The graphs below show data collected from the experiment, where the volume of hydrogen produced over time was recorded for different HCl concentrations.



- 2.6 Based on the graph, at which concentration does the reaction proceed the fastest? Explain your answer by referring to the gradient of each curve. (2)

- 2.7 Draw the expected graph for a 1,5 mol.dm⁻³ HCl solution. (2)

2.8 What is a *catalyst*, and how does it affect the rate of a reaction? (2)

2.9 If a catalyst were added to half of the volume of $2 \text{ mol.dm}^{-3} \text{ HCl}$, draw and label a new line on the graph to show the effect on the reaction rate. (2)

2.10 During the experiment, the student notices that the reaction rate decreases over time, even when the concentration of acid is high.

Explain why the rate of reaction generally decreases over time in this experiment. (3)

2.11 Explain why the downward displacement of water is an appropriate method for collecting hydrogen gas in this reaction. In your answer, consider the properties of hydrogen gas that make this method suitable. (2)

QUESTION 3**25 MARKS**

A student was asked to compare the industrial processes used to extract aluminium and iron from their ores.

The aluminium oxide reacts with hot concentrated sodium hydroxide and water to form a solution of sodium tetrahydroxoaluminate ($\text{NaAl}(\text{OH})_4$) according to the following balanced chemical equation:



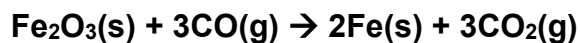
3.1 200 ml of a $2,0 \text{ mol}\cdot\text{dm}^{-3}$ solution of sodium hydroxide is reacted with excess aluminium oxide and 12 g of water:

3.1.1 Determine which reagent is limiting between the sodium hydroxide and the water. (4)

3.1.2 Explain the term “*yield*” in the context of a chemical reaction. (2)

3.1.3 If 45 g of $\text{NaAl}(\text{OH})_4$ is actually produced in this reaction, determine the percentage yield of $\text{NaAl}(\text{OH})_4$. (5)

In the extraction of iron from its ore, carbon monoxide is reacted with iron oxide according to the following balanced chemical equation:



- 3.2 A mass of 1 kg of Fe must be produced. Assume the reaction is 100% efficient and the carbon monoxide is in excess.

3.2.1 Determine the amount of iron oxide that is required. (3)

3.2.2 Determine the number of iron atoms that is **required**. (4)

3.2.3 Determine the volume of carbon dioxide that is produced if the reaction is carried out at STP. (3)

3.2.4 What mass of iron oxide would be required if the reaction was only 70% efficient? 1 kg of Fe must still be obtained. (4)

QUESTION 4

15 MARKS

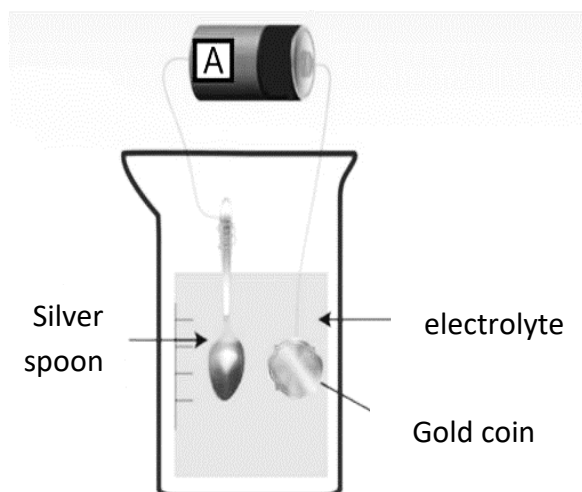
- 4.1 The entire world has watched the 2024 Olympics and marvelled at the beautifully designed medals that the French have designed.



The three types of medals have different metals.

The gold medal, largely includes silver with 6 grams of gold plating, [according to NPR](#).

The Grade 11s decides to experiment with metal electroplating and tests it out on a silver spoon.



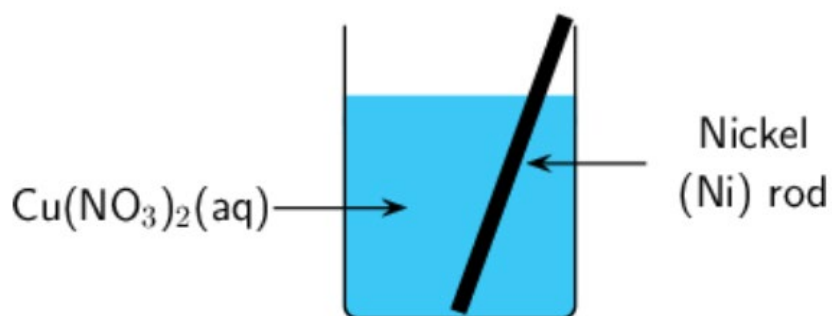
- 4.1.1 In order for the spoon to be electroplated, should **A** be the POSITIVE or NEGATIVE terminal of the cell? Circle your answer. (1)

- 4.1.2 Write the equation for the half-reaction taking place at the anode. (2)

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- 4.1.3 Define *electrolyte*. (2)

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- 4.1.4 Write the formula of an appropriate electrolyte of this experiment. (2)
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- 4.2 A nickel, Ni, rod is placed in a beaker containing a copper(II) nitrate solution, $\text{Cu}(\text{NO}_3)_2$, and a reaction takes place.



- 4.2.1 Write the dissociation equation for copper(II) nitrate. Include state symbols. (2)

- 4.2.2 Use oxidation number to explain which substance is oxidised and which is reduced. (4)

- 4.2.3 Write the balanced net ionic equation of the reaction. (2)

QUESTION 5**13 MARKS**

Consider the following organic compounds:

IUPAC Name	Molecular formula	Molar mass (g.mol ⁻¹)	Boiling point (°C)
Pentane	C ₅ H ₁₂	72	36
Dimethylpropane	C ₅ H ₁₂	72	10
Pent-1-ene	C ₅ H ₁₀	70	30
Butanoic acid	C ₄ H ₈ O ₂	88	163,5
Pentan-1-ol	C ₅ H ₁₂ O	88	138

- 5.1 Pentane and pent-1-ene are both colourless liquids at room temperature. Describe a chemical test to distinguish between the two liquids. (3)

- 5.2 Draw the structural formula of dimethylpropane. (2)

- 5.3 Dimethylpropane is a chain isomer of pentane.

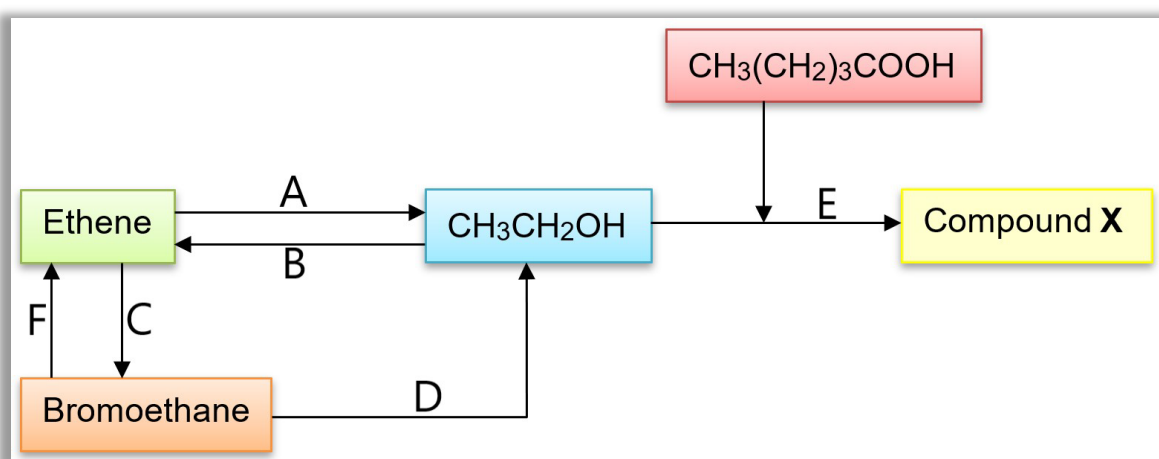
- 5.3.1 Define *isomers*. (2)

- 5.3.2 Draw the structural formula of third chain isomer of pentane. (2)

- 5.4 Explain why pentan-1-ol has a lower boiling point than butanoic acid, despite them having identical molar masses. (4)

QUESTION 6**34 MARKS**

The flow diagram below shows how ethene can be used to prepare various organic compounds. The letters **A** to **F** represent different organic reactions.



- 6.1 Define *homologous series*. (2)

- 6.2 To which homologous series does bromoethane belong? (1)

6.3 Define *functional group*. (2)

6.4 Consider compound $\text{CH}_3(\text{CH}_2)_3\text{COOH}$

6.4.1 Give the general formula for this compound. (1)

6.4.2 Draw the functional group of $\text{CH}_3(\text{CH}_2)_3\text{COOH}$. (2)

6.5 Identify the specific type of reaction represented by:

6.5.1 **B** _____ (1)

6.5.2 **D** _____ (1)

6.6 Write down TWO reaction conditions for reaction **B**. (2)

6.7 For reaction **A**, write down the:

6.7.1 name of the inorganic reactant. (1)

6.7.2 chemical formula of the catalyst needed. (1)

6.8 For reaction **C**:

6.8.1 Use structural formula and write down a balanced chemical equation. (3)

6.8.2 Explain why no water should be present during this reaction. (1)

6.9 Reaction **E** represents the conversion of the alcohol into organic compound **X**.
Write down the:

6.9.1 type of reaction. (1)

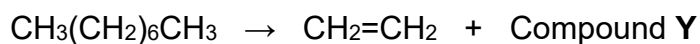
6.9.2 chemical formula of the catalyst needed. (1)

6.9.3 structural formula of compound **X**. (2)

6.9.4 IUPAC name of compound **X**. (2)

6.10 Reaction **F** takes place in the presence of warm, concentrated NaOH. Use condensed structural formula and write down a balanced equation for the reaction. (3)

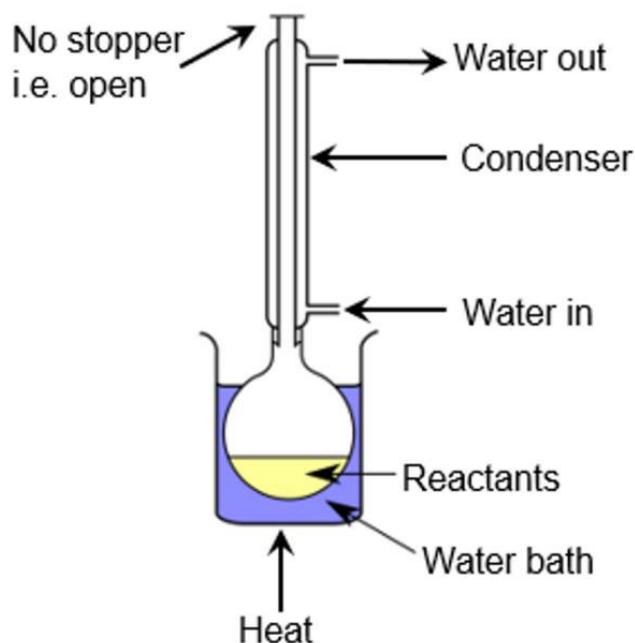
- 6.11 Large straight-chained alkanes can be catalytically cracked to produce shorter chained alkenes and branched alkanes which are more suitable for use in petrol. The reaction below indicates the catalytic cracking of octane.



- 6.11.1 Write down the IUPAC name of compound Y. (2)

- 6.11.2 Briefly explain why shorter-chained alkenes and branched alkanes are more suitable for use in petrol than large straight-chained alkanes. (2)

- 6.12 The diagram illustrates 'heating under reflux'.



[Image from IEB Paper 2 2021]

- 6.12.1 Which reaction (A – F) specifically uses 'heating under reflux'? (1)

- 6.12.2 Explain why reactions involving organic compounds are often conducted under reflux or in a water bath, and how these techniques contribute **specifically to effective reaction conditions**. (2)

[illegible]